

Lab Experiment: 4

Title: Playing Melody with Buzzer using Arduino

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Submitted to-

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Lab-4: Playing Melody with Buzzer using Arduino

1. Title Playing Melody with Buzzer using Arduino

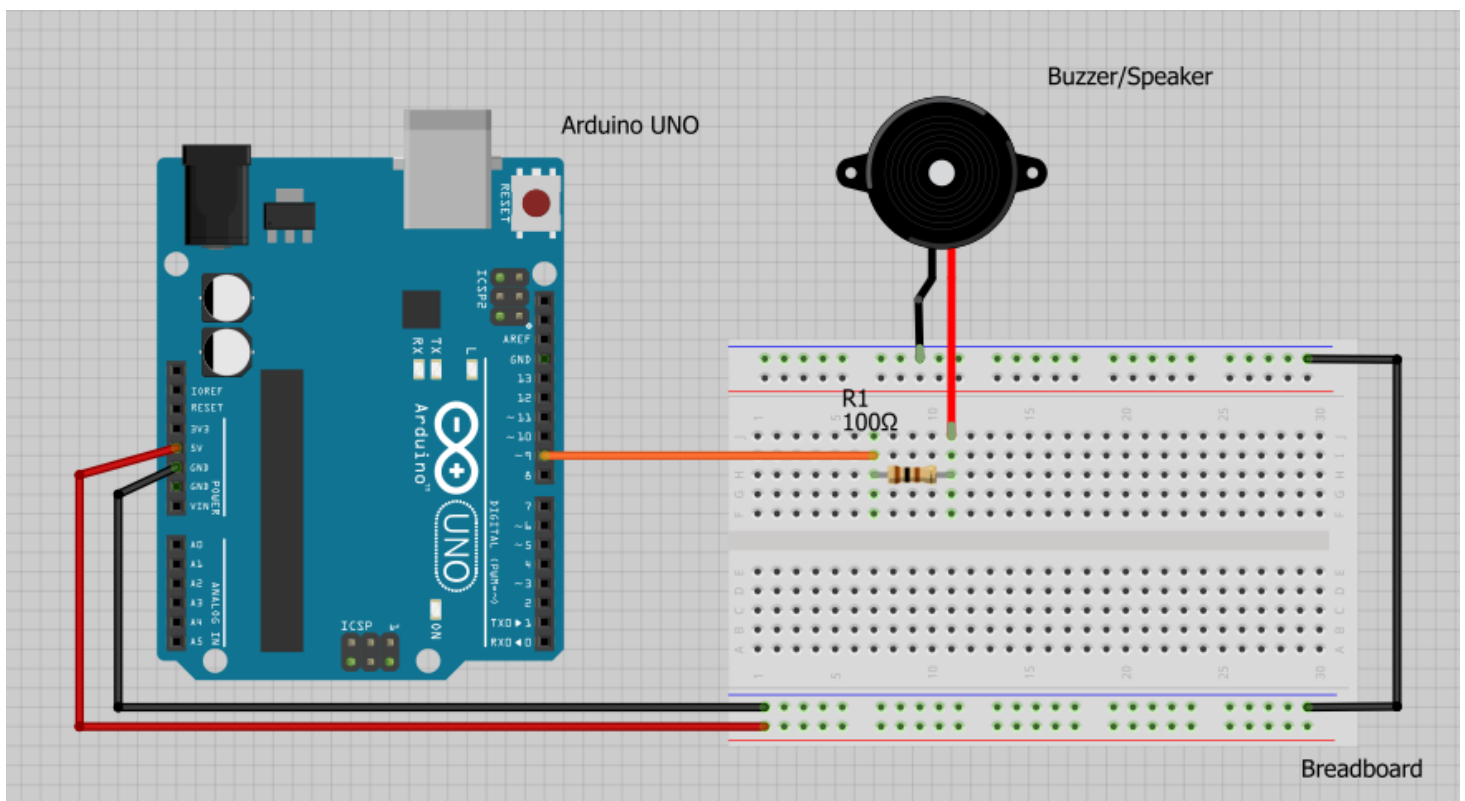
2. Objective

- To interface a buzzer with an Arduino board
- To generate sound using digital signals
- To play a simple musical melody using programmed notes
- To understand tone generation using frequency and delay

3. Components Required

Component	Quantity
Arduino Uno (or compatible board)	1
Buzzer (Active/Passive)	1
Jumper wires	As required
USB cable	1
Computer with Arduino IDE	1

4. Circuit Connection



- Buzzer **positive (+) terminal** → **Digital Pin 9** of Arduino
- Buzzer **negative (-) terminal** → **GND** of Arduino

This allows Arduino to send HIGH/LOW signals to produce sound waves.

5. Program (Arduino Code)

```
int buzzerpin = 9;

int length = 15;
char notes[] = "ccggaagffeeddc ";
int beats[] = {1,1,1,1,1,1,2,1,1,1,1,1,1,2,4};
int tempo = 300;

void playTone(int tone, int duration) {
    for (long i = 0; i < duration * 1000L; i += tone * 2) {
        digitalWrite(buzzerpin, HIGH);
        delayMicroseconds(tone);
        digitalWrite(buzzerpin, LOW);
        delayMicroseconds(tone);
    }
}

void playNote(char note, int duration) {
    char names[] = {'c','d','e','f','g','a','b','C'};
    int tones[] = {1915,1700,1519,1432,1275,1136,1014,956};

    for (int i = 0; i < 8; i++) {
        if (names[i] == note) {
            playTone(tones[i], duration);
        }
    }
}

void setup() {
    pinMode(buzzerpin, OUTPUT);
}

void loop() {
    for (int i = 0; i < length; i++) {
        if (notes[i] == ' ') {
            delay(beats[i] * tempo);
        } else {
            playNote(notes[i], beats[i] * tempo);
        }
        delay(tempo / 2);
    }
}
```

6. Working Principle

- Arduino outputs square wave signals from Pin 9
- The buzzer converts electrical pulses into sound
- Different frequencies correspond to different musical notes
- Delay determines note duration (tempo)

- A sequence of notes forms a melody

7. Procedure

1. Collect all required components
2. Connect buzzer to Arduino as shown
3. Open Arduino IDE on the computer
4. Write or paste the provided code
5. Select correct Board and COM port
6. Upload the program to Arduino
7. Observe the buzzer playing the melody

8. Result

The buzzer successfully produced a musical melody based on programmed notes, demonstrating sound generation using Arduino digital output.

9. Applications

- Alarm systems
- Doorbells
- Musical toys
- IoT notification devices
- Warning indicators in embedded systems

10. Precautions

- Ensure correct polarity of buzzer
- Avoid short circuits
- Use proper digital pin connection
- Check board and port before uploading

11. Conclusion

This experiment demonstrated how an Arduino microcontroller can generate sound and play melodies using a buzzer. It provided practical understanding of digital signal output, timing control, and basic sound synthesis — fundamental concepts in IoT and embedded systems.